

Torcetrapib Retrospective Pilot — Case Study Summary

AgenThink Mesh | Pharma Council V1 | Retrospective Validation Session ID: PHARMA-RETRO-TORCETRAPIB-1781180831427 Report ID: IPR-PHARMA-RETRO-TORCETRAPIB-2026-06-11 Generated: 2026-06-11

THE BET

For torcetrapib to succeed in Phase III, the +2 mmHg systolic BP increase observed in Phase II must be either (a) a statistical artifact, (b) a manageable class effect shared by all CETP inhibitors, or © offset by the HDL-C benefit in the high-CV-risk population. None of these conditions were established before Pfizer advanced.

Verdict

WAIT — 0 GO / 8 WAIT / 2 NO-GO — Proof Score: $\frac{93}{100}$

10 of 10 council members voted WAIT or NO-GO. The primary blocker is the unresolved systolic blood pressure signal (+2 mmHg) in the Phase II dataset, which requires mechanistic investigation before enrolling 15,000 high-cardiovascular-risk patients in Phase III.

3 Reasons the Council Said WAIT / NO-GO

1. Unresolved BP Signal — No Mechanistic Explanation Brousseau et al. NEJM 2004 (Table 3) documented a statistically significant +2 mmHg systolic BP increase at the 60mg dose. Pfizer characterized it as “small but statistically significant.” The council applied Lewington et al. Lancet 2002: each 2 mmHg systolic BP increase = 7% stroke mortality increase + 4% ischemic heart disease mortality increase. In a 15,000-patient high-CV-risk Phase III population, this is not a small signal — it is the primary outcome driver.

2. Molecule-Specific Signal — Not a Class Effect No competing CETP inhibitor (dalcetrapib, anacetrapib) showed a comparable BP signal in available Phase I/II data as of December 31, 2005. This means the BP effect is likely torcetrapib-specific, not a CETP class effect. A molecule-specific unexplained BP increase in a cardiovascular drug is a hard blocker under PC-007 (Mechanistic Investigation Requirement).

3. Surrogate Endpoint Risk — HDL-C Not Validated for CV Benefit HDL-C increase is a surrogate endpoint. The validated pathway from HDL-C increase to cardiovascular event

reduction is not established. If the BP increase offsets or reverses the HDL benefit, the trial will show net harm — which is exactly what ILLUMINATE demonstrated.

What Would Change the Decision

The council identified four conditions that would allow advancement to Phase III:

Requirement	Rule	Status
Mechanistic study: is BP increase on-target (CETP) or off-target (torcetrapib-specific)?	PC-007	Not completed before Phase III
DSMB charter amendment: explicit BP stopping rules	PC-008	Not in ILLUMINATE charter
Competitive class comparison: BP data from dalcetrapib and anacetrapib	PC-005	Not obtained before Phase III
Quantitative risk-benefit analysis: BP harm vs HDL benefit in Phase III population	PC-006	Not commissioned before Phase III

None of these conditions were met before Pfizer advanced torcetrapib to Phase III.

Council Vote Distribution

Persona	Vote	Confidence	Primary Flag
Chief Biostatistician	WAIT	95%	BP signal mechanistic investigation pending
Clinical Pharmacologist	WAIT	85%	Competitive landscape BP signal discrepancy
Regulatory Strategist	WAIT	95%	Unvalidated surrogate-to-clinical pathway
Drug Safety Expert	WAIT	95%	Unexplained BP signal
Portfolio Manager	WAIT	90%	Potential off-target effect
Scientific Skeptic	WAIT	95%	Lack of mechanistic understanding
Commercial Assessor	WAIT	85%	Uninvestigated BP signal; competitive landscape BP comparison
Patient Advocate	NO-GO	95%	DSMB charter inadequate
Quality / Compliance Expert	WAIT	95%	Gross safety signal; off-target cardiovascular effect
Devil's Advocate	NO-GO	95%	Off-target cardiovascular effect

Retrospective Validation

The council verdict was retrospectively correct.

Event	Date	Data
ILLUMINATE trial termination	December 2, 2006	DSMB terminated for excess mortality
Deaths	—	82 (torcetrapib arm) vs 51 (control arm)
BP finding in Phase III	—	+5.4 mmHg systolic (vs +2 mmHg in Phase II — the signal grew)
R&D write-off	December 3, 2006	\$800M
Pfizer market cap loss	December 3, 2006	\$21B
Off-target mechanism confirmed	2008	Forrest et al. J Pharmacol Exp Ther: aldosterone secretion stimulation

The Forrest et al. 2008 publication confirmed that torcetrapib stimulates aldosterone secretion via an off-target mechanism independent of CETP inhibition. This is exactly the molecule-specific off-target effect the council flagged as the primary blocker. A mechanistic investigation before Phase III would have identified this mechanism and prevented the trial.

Evidence Boundary Statement

EVIDENCE BOUNDARY CONFIRMED: All evidence provided to the council is dated on or before December 31, 2005. The ILLUMINATE Phase III trial results (December 2, 2006), the 82 vs 51 death finding, the \$800M write-off, the market cap loss, and the Forrest et al. 2008 off-target mechanism confirmation are **EXCLUDED** from council input. These post-failure data points appear only in the Retrospective Outcome Appendix of the Institutional Proof Report (Section 15), clearly separated from the council deliberation record.

What This Proves

This pilot demonstrates that AgenThink Mesh's governed council infrastructure, operating on publicly available pre-failure evidence, would have issued a WAIT verdict on torcetrapib before Pfizer advanced to Phase III. The primary blocker — the unresolved BP signal — was the exact mechanism that caused 82 deaths and an \$800M write-off 18 months later.

The system did not need post-failure knowledge to reach the correct verdict. It needed only:

1. The Phase II BP data (publicly available in NEJM 2004)
2. The epidemiological risk quantification (publicly available in Lancet 2002)
3. The competitive class comparison (available from dalcetrapib/anacetrapib Phase I/II programs)
4. A constitutional rule requiring mechanistic investigation before Phase III advancement

All four were available. None were applied by Pfizer's internal process. The council applied all four.

Next Steps

1. **Publish this case study** as a white paper for pharma R&D leadership
2. **Run the next retrospective pilot** (recommended: aducanumab or verubecestat)
3. **Build the prospective pharma module** using this engine as the validated foundation
4. **Use the Institutional Proof Report** as the primary demonstration asset for pharma clients

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